# Lab 05 - Diagnose Pipeline Flow and Cultural Debt

Course: DevOps Foundation Training (TGS-2024049184) Version: v4.0 Topic: 05 Estimated time: 35-45 minutes

## Workplace scenario

Teams blame the test stage for slow delivery, but no one has quantified queue time, rework or manual hand-offs across the pipeline.

## Objective

Apply K4 and A4 by combining stage performance with visible queue and rework evidence to recommend a prioritised improvement.

## What you will produce

A pipeline_stage_summary.csv, pipeline_flow.png and a generative-culture recommendation.

## Prerequisites

- R 4.x available as Rscript.
- Work from this lab folder so relative paths resolve.
- Use only the supplied synthetic data; no internet connection or external package is required.
- Preserve the source data. Make experimental changes in a copied file, then restore it.

## Folder map

```
Lab-05-pipeline-risk-review/
■■■ README.md / README.pdf
■■■ CHECKLIST.md / CHECKLIST.pdf
■■■ pipeline_risk_review.R
■■■ data/
■■■ output/            # reproducible evidence produced by the R script
```

## Detailed procedure

### Step 1 - Inspect pipeline_runs.csv and identify active time, wait time, failures and rework.

```
Rscript -e 'd <- read.csv("data/pipeline_runs.csv"); str(d)'
```

Record what you observed before continuing.

### Step 2 - Run pipeline_reliability.R to calculate stage-level flow and instability indicators.

```
Rscript pipeline_reliability.R
```

Record what you observed before continuing.

### Step 3 - Review pipeline_stage_summary.csv and identify the current constraint using evidence rather than reputation.

```
Rscript -e 'print(read.csv("output/pipeline_stage_summary.csv"))'
```

Record what you observed before continuing.

### Step 4 - Open pipeline_flow.png and locate the stage with the largest wait-to-active ratio.

```
open output/pipeline_flow.png
```

Record what you observed before continuing.

### Step 5 - Translate the finding into one CALMS improvement and one generative-culture behaviour.

```
open CHECKLIST.md
```

Record what you observed before continuing.

### Step 6 - Record a success metric and 30-day review date for the recommendation.

```
open CHECKLIST.md
```

Record what you observed before continuing.

## Evidence to retain

[ ] output/pipeline_stage_summary.csv

[ ] output/pipeline_flow.png

[ ] completed CHECKLIST.md

## Acceptance criteria

- Constraint identified from data
- No individual-blame language
- CALMS dimension named
- Metric and review date recorded

## Verification

The identified constraint matches the maximum wait ratio; the recommendation changes a system condition and names an outcome measure.

## Troubleshooting

- cannot open file: confirm your terminal is in this lab folder.
- Rscript: command not found: install R, reopen the terminal and run Rscript --version.

- A validation stop(...) message is evidence that the input broke a stated rule; inspect the named field rather than bypassing the check.
- If an output looks wrong, compare row counts and IDs with the source CSV before changing the analysis.

## Clean rerun

Delete only the files inside output/, then run the R script again. The source CSV and script remain unchanged.